

CHAPTER FOUR

CHI-SQUARE DISTRIBUTIONS

4.1. Basic Concepts

A Chi-square (χ^2) distribution is a continuous distribution ordinarily derived as the sampling distribution of a sum of squares of independent standard normal variables.

Characteristics of the square distributions

- It is a continuous distribution
- The χ^2 dist has a single parameter; the degree of freedom, v
- The mean of the chi-square distribution is v
- The variance of the chi-square distribution is $2v$. Thus, the mean and Variance depend on the degree of freedom.
- It is based on a comparison of the sample of observed data (results) with the expected results under the assumption that the null hypothesis is true.
- It is a skewed distribution and only nonnegative values of the variable χ^2 are possible. The skewness decreases as v increases; and when V increases without limit it approaches a normal distribution. It extends indefinitely in the positive direction
- The area under the curve is 1.0

Having the above characteristics, χ^2 distribution has the following areas of application:

- Test for independence between two variables
- Testing for the equality of several proportions
- Goodness of fit tests (Binomial, Normal, and Poisson)

4.2. Test for the Independence between Two Variables

A χ^2 test of independence is used to analyze the frequencies of two variables with multiple categories to determine whether the two variables are independent. That is, the Chi-square distribution involves using sample data to test for the independence of two variables. The sample data are given in to a two way table called a contingency table. Because the χ^2 test of independence uses a contingency table, the test is sometimes referred to as **contingency analysis** (Contingency table test). The χ^2 test is used to analyze, for example, the following cases:

- Whether employee absenteeism is independent of job classification
- Whether beer preference is independent of sex (gender)
- Whether favorite sport is independent of nationality.
- Whether type of financial investment is independent of geographic region.

The steps and procedures are similar with hypothesis testing. The Chi-square test for independence is useful in helping to determine whether a relationship exists between two variables, but it does not enable us to estimate or predict the values of one variable based on the value of the other. If it is determined that a dependence does exist between two quantitative variables, then the techniques of regression analysis are useful in helping to find a mathematical formula that expresses the nature of mathematical relationship.

Small expected frequencies can lead to inordinately large chi-square values with the chi-square test of independence. Hence contingency tables should not be used with expected cell values of less than 5.

One way to avoid small expected values is to combine columns or rows whenever possible and whenever doing so makes sense.

Illustration One

A company planning a TV advertising campaign wants to determine which TV shows its target audience watches and thereby to know whether the choice of TV program an individual watches is independent of the individual's income. The table supporting this is shown below. Use a 5% level of significance and the null hypothesis.

Income	Type of Show			
	Basketball	Movie	News	Total
Low	143	70	37	250
Medium	90	67	43	200
High	17	13	20	50
Total	250	150	100	500

Solution

Step 1. Ho: Choice of TV program an individual watches is independent of the individual's income

Ha: Income and Choice of TV program are not independent

Step 2. Decision rule

$$\alpha = 0.05$$

$$\begin{aligned} v &= (R-1)(C-1)^1 \\ &= (3-1)(3-1) \\ &= 4 \end{aligned}$$

$$\chi^2_{\alpha, v} = \chi^2_{0.05, 4} = 9.49$$

Reject Ho if sample χ^2 is greater than 9.49

Step 3. Compute the test statistic

In computing the test statistic our first task is to estimate the expected frequencies ($e_{ij} = r_i c_j / n$); where

r_i = Observed freq total for row i.

c_j = observed freq total for column j

n = sample size

$e_{11} = 250 \times 250 / 500 = 125$	$e_{21} = 200 \times 250 / 500 = 100$	$e_{31} = 50 \times 250 / 500 = 25$
$e_{12} = 250 \times 150 / 500 = 75$	$e_{22} = 200 \times 150 / 500 = 60$	$e_{32} = 50 \times 150 / 500 = 15$
$e_{13} = 250 \times 100 / 500 = 50$	$e_{23} = 200 \times 100 / 500 = 40$	$e_{33} = 50 \times 100 / 500 = 10$

A test of the null hypothesis that variables are independent of one another is based on the magnitudes of the differences between the observed frequencies and the expected frequencies. Large differences between o_{ij} and e_{ij} provide evidence that the null hypothesis is false. The test is based on the following Chi-square test statistic.

¹For the RxC contingency table (R=no. of Rows and C = Columns), the degrees of freedom are calculated as (R-1)(C-1). The degrees of freedom refers to the number of expected frequencies that can be chosen freely provided the row and column totals of expected frequencies are identical to the row and column totals of the observed frequency table.

$$\chi^2 = \sum \frac{(O_{ij} - e_{ij})^2}{e_{ij}} \quad \text{Or} \quad \chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

Where:

O_{ij} (f_o) = observed frequency for contingency table category in row i and column j.

e_{ij} (f_e) = expected frequency for contingency table in row i and column j.

$$\chi^2 = \frac{(143-125)^2}{125} + \frac{(70-75)^2}{75} + \frac{(37-50)^2}{50} + \frac{(90-100)^2}{100} + \frac{(67-60)^2}{60} + \frac{(17-25)^2}{25} + \frac{(13-15)^2}{15} + \frac{(20-10)^2}{10} + \frac{(43-40)^2}{40} = 21.174$$

Step 4. Reject the null hypothesis that choice of TV program is independent from income level.

Illustration Two

A human resource manager at EAGLE Inc. was interested in knowing whether the voluntary absence behavior of the firm's employees was independent of marital status. The employee files contained data on marital status and on voluntary absenteeism behavior for a sample of 500 employees is shown below.

Marital Status					
Absence behavior	Married	Divorced	Widowed	Single	Total
Often absent	36	16	14	34	100
Seldom absent	64	34	20	82	200
Never absent	50	50	16	84	200
Total	150	100	50	200	500

Test the hypothesis that absence behavior is independent of marital status at a significance level of 1%.

Solution

Step 1. H_o : Voluntary absence behavior is independent of marital status

H_a : Voluntary absence behavior and marital status are dependent

Step 2. $\alpha = 0.01$

$$V = (R-1)(C-1)$$

$$= (3-1)(4-1) = 6$$

$$\chi^2_{\alpha, v} = \chi^2_{0.01, 6} = 16.81$$

Reject H_o if sample $\chi^2 > 16.81$

Step 3. Sample χ^2

Observed freq(f_o)	Expected Freq (f_e)	$(f_o - f_e)^2$	$\frac{(f_o - f_e)^2}{f_e}$
36	30	36	1.200
64	60	16	0.267
50	60	100	1.667

16	20	16	0.800
34	40	36	0.900
50	40	100	2.500
14	10	16	1.600
20	20	0	0.000
16	20	16	0.800
34	40	36	0.900
82	80	4	0.050
84	80	16	0.200
$\sum \frac{(f_o - f_e)^2}{f_e}$			10.883

Step 4. Do not reject H_0 ; because 10.883 is less than 16.81. Thus, voluntary absence and marital status are independent.

Illustration Three

The personnel administrator of XYZ Company provided the following data as an example of selection among 40 male and 40 female applicants for 12 open positions.

Applicant	Status		
	Selected	Not selected	Total
Male	7	33	40
Female	5	35	40
Total	12	68	80

- The χ^2 test of independence was suggested as a way of determining if the decision to hire 7 males and females should be interpreted as having a selection bias in favor of males. Conduct the test of independence using $\alpha = 0.10$. What is your conclusion?
- Using the same test, would the decision to hire 8 males and 4 females suggest concern for a selection bias?

Solutions

a. **Step 1.** H_0 : There is no selection bias in favor of males. (Selection status and gender of the applicant are independent).

H_a : There is selection bias in favor of males. (Selection status and gender of the applicant are not independent).

Step 2. $\alpha = 0.1$

$$\begin{aligned}
 V &= (R-1)(C-1) \\
 &= (2-1)(2-1) = 1 \\
 \chi^2_{\alpha, V} &= \chi^2_{0.1, 1} = 2.71
 \end{aligned}$$

Reject H_0 if sample $\chi^2 > 2.71$

Step 3. Sample χ^2

Observed freq (f_o)	Expected Freq (f_e)	($f_o - f_e$) ²	$\frac{(f_o - f_e)^2}{f_e}$
7	6	1	0.1667
33	34	1	0.0294
5	6	1	0.1667
35	34	1	0.0294
$\sum \frac{(f_o - f_e)^2}{f_e}$			0.3922

Step 4. Do not reject H_0 ; because 0.392 is less than 2.71. Thus, there is no selection bias in favor of male applicants.

b. Step 1. H_0 : There is no selection bias in favor of males. (Selection status and gender of the applicant are independent).

H_a : There is selection bias in favor of males. (Selection status and gender of the applicant are not independent).

Step 2. $\alpha = 0.1$

$$V = (R-1)(C-1) \\ = (2-1)(2-1) = 1$$

$$\chi^2_{\alpha, v} = \chi^2_{0.1, 1} = 2.71$$

Reject H_0 if sample $\chi^2 > 2.71$

Step 3 Sample χ^2

Observed freq (f_o)	Expected Freq (f_e)	($f_o - f_e$) ²	$\frac{(f_o - f_e)^2}{f_e}$
8	6	4	0.6667
32	34	4	0.1176
4	6	4	0.6667
36	34	4	0.1176
$\sum \frac{(f_o - f_e)^2}{f_e}$			1.5686

Step 4 . Do not reject H_0 ; because 1.569 is less than 2.71. Therefore, there is no selection bias in favor of male applicants.

4.3. Testing For the Equality of Several Proportions

Testing for the equality of several proportions emphasizes on whether several proportions are equal or not; and hence the null hypothesis takes the following form:

$$H_0: P_1 = P_{10}; P_2 = P_{20}; P_3 = P_{30}; \dots P_k = P_{k0};$$

And the alternative hypothesis takes the following form:

H_a : The population proportions are not equal to the hypothesized values

The degree of freedom is determined as $V = K - 1$; where K refers to the number of proportions and all expected cell values must be greater than or equal to 5.

Illustration One

In the business credit institution industry the accounts receivable for companies are classified as being “current,” “moderately late,” “very late,” and “uncollectible.” Industry figure show that the ratio of these four classes is 9: 3: 3: 1. ENDURANCE firm has 800 accounts receivable, with 439, 168, 133, and 60 falling in each class. Are these proportions in agreement with the industry ratio? Let $\alpha=0.05$.

Solution

Step 1. Ho: $P_1 = 9/16$; $P_2 = 3/16$; $P_3 = 3/16$; $P_4 = 1/16$

Ha: One or more of the proportions are not equal to the proportions given in the null hypothesis.

Step 2. $\alpha = 0.05$

$$v = K - 1 = 4 - 1 = 3$$

$$\chi^2_{\alpha, v} = \chi^2_{0.05, 3} = 7.81$$

Reject Ho if sample $\chi^2 > 7.81$

Step 3. Test Statistics (Sample χ^2)

Class	Observed freq (f_o)	Expected Freq ($f_e = np_i$)	$(f_o - f_e)^2$	$\frac{(f_o - f_e)^2}{f_e}$
Current	439	450	121	0.269
Moderately late	168	150	324	2.160
Very late	133	150	289	1.927
Uncollectible	60	50	100	2.000
$\sum \frac{(f_o - f_e)^2}{f_e}$				6.356

Step 4. Since static value (7.81) greater than the estimated value (6.356), do not reject.

Illustration ETHIO Plastic Factory sells its products in three primary colors: Red, blue, and yellow. The marketing manager feels that customers have no color preference for the product. To test this hypothesis the manager set up a test in which 120 purchases were given equal opportunity to buy the product in each of the three colors. The results were that 60 bought red, 20 bought blue, and 40 bought yellow. Test the marketing manager’s null hypothesis, using $\alpha=0.05$.

Solution

Step 1. Ho: People have no color preference with this product; $P_1 = P_2 = P_3 = 1/3$

Ha: People have color preference with this product

Step 2. $\alpha = 0.05$

$$V = K - 1 = 3 - 1 = 2$$

$$\chi^2_{\alpha, v} = \chi^2_{0.05, 2} = 5.99$$

Reject Ho if sample χ^2 is greater than 5.99.

Step 3. Sample χ^2

	Observed freq (f_o)	Expected Freq ($f_e = np_i$); $p_i = 1/3^2$	$(f_o - f_e)^2$	$\frac{(f_o - f_e)^2}{f_e}$
Red	60	40	400	10.00
Blue	20	40	400	10.00
Yellow	40	40	0	0.00
$\sum \frac{(f_o - f_e)^2}{f_e}$				20.00

Step 4. Reject H_0 ; because $20 > 5.99$. This means that customers do have color preference. It appears that red is the most popular color and blue is the least popular.

Illustration Three

Rating sciences, Inc., a TV program – rating service, surveyed 600 families where the television was turned on during the prime time on week nights. They found the following numbers of people turned to the various networks.

Name of the network	Type	Number of viewers
NBC	Commercial	210
CBS		170
ABC		165
PBS	Noncommercial	55
Total		600

- Test the hypothesis that all four networks have the same proportion of viewers during this prime time period. Use $\alpha = 0.05$
- Eliminate the results for PBS and repeat the test of hypothesis for the three commercial networks, using $\alpha = 0.05$
- Test the hypothesis that each of the three major networks has 30% of the weeknight prime time market and PBS has 10% using $\alpha = 0.005$

Solutions

- a) Step 1.** H_0 : All of the four networks do have equal number of viewers; $P_1 = P_2 = P_3 = P_4 = 1/4$.
 H_a : All of the four networks do not have equal number of viewers.

Step 2. $\alpha = 0.05$

$$V = K - 1 = 4 - 1 = 3$$

$$\chi^2_{\alpha, V} = \chi^2_{0.05, 3} = 7.81$$

Reject H_0 if sample χ^2 is greater than 7.81

Step 3. Sample χ^2

²Since the null hypothesis states that there is no color preference, each of the three colors is preferred by one third of the purchases.

Class	Observed freq (f_o)	Expected Freq ($f_e = np_i$); $p_i = 1/4$ ³	($f_o - f_e$) ²	$\frac{(f_o - f_e)^2}{f_e}$
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³Since equal numbers of viewers are expected to watch each network, each of the four networks is watched by one fourth of the viewers.

NBC	210	150	3,600	24.0000
CBS	170	150	400	2.6667
ABC	165	150	225	1.5000
PBS	55	150	9,025	60.1667
$\sum \frac{(f_o - f_e)^2}{f_e}$				88.3334

Step 4. Reject H_0 ; because $88.34 > 7.81$.

b) Step 1. H_0 : All of the three commercial networks do have equal number of viewers; $P_1 = P_2 = P_3 = 1/3$.

H_a : All of the three commercial networks do not have equal number of viewers.

Step 2. $\alpha = 0.05$

$$V = K - 1 = 3 - 1 = 2$$

$$\chi^2_{\alpha, V} = \chi^2_{0.05, 2} = 5.99$$

Reject H_0 if sample χ^2 is greater than 5.99.

Step 3. Sample χ^2

Class	Observed freq (f_o)	Expected Freq ($f_e = np_i$); $p_i = 1/3$	$(f_o - f_e)^2$	$\frac{(f_o - f_e)^2}{f_e}$
NBC	210	181.67	802.60	4.4179
CBS	170	181.67	136.20	0.7497
ABC	165	181.67	277.90	1.5270
$\sum \frac{(f_o - f_e)^2}{f_e}$				6.6946

Step 4. Reject H_0 ; because $6.70 > 5.99$

c) Step 1. H_0 : $P_1 = P_2 = P_3 = 0.30$; $P_4 = 0.10$

H_a : One or more of the proportions are not equal to the proportions given in the null hypothesis.

Step 2. $\alpha = 0.005$

$$V = K - 1 = 4 - 1 = 3$$

$$\chi^2_{\alpha, V} = \chi^2_{0.005, 3} = 7.81$$

Reject H_0 if sample χ^2 is greater than 7.81

Step 3. Sample χ^2

Class	Observed freq (f_o)	Expected Freq ($f_e = np_i$)	$(f_o - f_e)^2$	$\frac{(f_o - f_e)^2}{f_e}$
NBC	210	180	900	5.00
CBS	170	180	100	0.55
ABC	165	180	225	1.25

PBS	55	60	25	0.42
$\sum \frac{(f_o - f_e)^2}{f_e}$				7.22

Step 4. Do not Reject H_0 ; because $7.22 < 7.81$.

Illustration 4

Suppose that three companies, A, B, and C, have recently conducted aggressive advertising campaigns in order to maintain and possibly increase their respective shares of the market for a particular product. The market shares prior to the campaigns were $P_1=0.45$ for company A, $P_2 = 0.40$ for company B, $P_3=0.13$ for company C, and $P_4 = 0.02$ for other competitors. To determine if these market shares changed after the advertising campaigns, a marketing analyst solicited the preferences of a random sample of 200 customers of this product. Of these 200 customers 95 indicated a preference for company A's product, 85 preferred company B's product, 18 preferred company C's product, and the remainder preferred one or another of the products distributed by the competitors conduct a test, at the 5% level of significance. If the market shares have changed from the levels what they were at before the advertising campaigns.

Solution

Step 1. $H_0: P_1 = 0.45; P_2 = 0.40; P_3 = 0.13; P_4 = 0.02$

H_a : At least one P_i is not equal to its specified value

Step 2. $\alpha = 0.05$

$$V = K - 1 = 3 - 1 = 2^4$$

$$\chi^2_{\alpha, v} = \chi^2_{0.05, 2} = 5.99$$

Reject H_0 if sample $\chi^2 > 5.99$.

Because of the above change H_0 is restated as:

$H_0: P_1 = 0.45, P_2=0.40, P_3 = 0.15$

H_a : At least one p_i is not equal to its specified value

Step 3. Sample X^2

Company	Observed freq (f_o)	Expected freq (f_e)	$(f_o - f_e)^2$	$\frac{(f_o - f_e)^2}{f_e}$
A	95	90	25	0.2778
B	85	80	25	0.125
Others	20	30	100	3.3333
$\sum \frac{(f_o - f_e)^2}{f_e}$				3.9236

Step 4. Do not reject H_0 . Thus there is no sufficient evidence at the 5% level of significance to conclude that the market shares have changed from the levels they were at before the advertising campaign.

⁴ Expected frequency value for P_4 is less than 5 ($200 \cdot 0.02 = 4$). So, we have to combine P_4 with one of other expected frequencies, say P_3 , to obtain a combined expected frequency of 30 ($200 \cdot 0.15$). it can also be combined with other expected frequency values.

4.4. Goodness-of-Fit Tests

The chi-square test is widely used for a variety of analyses. One of the more important uses of Chi-Square is the goodness-of-fit test. That is, it can be used to decide whether a particular probability distribution, such as the binomial, Poisson or normal, is the appropriate distribution. This is an important ability, because as decision makers using statistics, we will need to choose a certain probability distribution to represent the distribution of the data we happen to be considering.

In tests of hypothesis (Chapter 3), we assumed that the population was normal and tested the hypothesis $\mu = \mu_0$, $p = p_0$, etc. But what if we want to check on the assumption of normality itself? The multinomial χ^2 goodness-of-fit test can be applied.

The null hypothesis for a goodness-of-fit test is that the distribution of the population from which a sample is taken is the one specified. The alternative hypothesis is that the actual distribution is not the specified distribution. Generally, a researcher specifies only the name of distribution and uses the sample data to estimate the particular parameters of the distribution. In this situation one degree of freedom is lost for each parameter that has to be estimated. However, if the research completely specifies the distribution including parameter values, then no additional degrees of freedom is lost.

Null hypothesis	Parameters to be estimated	Degrees of freedom lost
Ho: Population is normal	μ, σ	2
Ho: Population is normal with $\mu = x$	σ	1
Ho: Population is normal with $\sigma = y$	μ	1
Ho: Population is normal with $\mu = x$, $\sigma = y$	None	0
Ho: Population is Poisson	λ	1
Ho: Population is Poisson with $\lambda = Z$	None	0
Ho: Population is binomial with $p = b$	None	0

4.4.1. Goodness-of-Fit Test for Binomial Distribution

Illustration One

Mrs. Tsion, Saleswoman for MOON Paper Company, has five accounts to visit per day. It is suggested that sales by Mrs. Tsion may be described by the binomial distribution, with the probability of selling each account being 0.4. Given the following frequency distribution of Mrs. Tsion's number of sales per day, can we conclude that the data do in fact follow the binomial distribution? Use the 0.05 significance level.

No. of sales day	0	1	2	3	4	5
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Frequency	10	41	60	20	6	3

Solution

Step 1. Ho: The frequency distribution is Binomial with $n = 5$ and $P = 0.4$

Ha: The frequency distribution is not binomial with $n = 5$ and $P = 0.4$

Step 2. $\alpha = 0.05$

$$K-1 - m = 5-1-0 = 4$$

$$\chi^2_{\alpha, v} = \chi^2_{0.05, 4} = 9.49$$

Reject Ho if sample χ^2 is greater than 9.49

Step 3. Sample χ^2 .

No. of sales per day	Prob. with $n=5$, $p=0.4$	Observed freq (f_o)	Expected Freq ($f_e = np_i$)	$(f_o - f_e)^2$	$\frac{(f_o - f_e)^2}{f_e}$
0	.0778	10	10.892	0.7957	0.0731
1	.2592	41	36.288	22.2029	0.6119
2	.3456	60	48.384	134.9315	2.7888
3	.2304	20	32.256	150.2095	4.6567
4&5	.0870	9	12.18	10.1124	0.8302
$\sum \frac{(f_o - f_e)^2}{f_e}$					8.9607

Step 4. Do not reject Ho. The data are well described by the binomial distribution with $n=5$ and $P=0.4$.

Illustration Two

A professional baseball player, Philippos, was at bat five times in each of 100 games. Philippos claims that he has a probability of 0.4 of getting a hit each time he goes to bat. Test his claim at the 0.05 level by seeing if the following data are distributed binomially.

No. of hits / game	0	1	2	3	4	5
No. of games with that number of hits	12	38	27	17	5	1

Solution

Step 1. Ho: The freq. Distribution can be best described by binomial distribution with $n=5$, $P=0.4$

Ha: The freq. Distribution can't be best described by binomial distribution with $n=5$, $P=0.4$

Step 2. $\alpha = 0.05$

$$V = K-1 - m = 5-1-0 = 4$$

$$\chi^2_{\alpha, v} = \chi^2_{0.05, 4} = 9.49$$

Reject H_0 if sample $\chi^2 > 9.49$

Step 3. Sample χ^2

No. of hits per game	No. of games with that no. of hit (f_o)	Prob. with $n=5$, $P=0.4$	Expected freq ($f_e = np_i$)	$(f_o - f_e)^2$	$\frac{(f_o - f_e)^2}{f_e}$
0	12	.0778	7.78	17.8084	2.2890
1	38	.2592	25.92	145.9264	5.6249
2	27	.3456	34.56	57.1536	1.6538
3	17	.2304	23.04	36.4816	1.5834
4 & 5	6	.0870	8.70	4.2900	0.8379
$\sum \frac{(f_o - f_e)^2}{f_e}$					11.9940

Step 4. Reject H_0 . The data is not binomially distributed with $n=5$ and $p=0.4$

Illustration Three

The Ethiopian postal service is interested in modeling the “mangled letter” problem. It has been suggested that any letter sent to a certain area has a 0.15 chance of being mangled. Since the post office is so big, it can be assumed that two letters chances of being mangled are independent. A sample of 310 people was selected, and two test letters were mailed to each of them. The number of people receiving zero, one, or two mangled letters was 260, 40, and 10, respectively. At the 0.10 level of significance, is it reasonable to conclude that the number of mangled letters received by people follows a binomial distribution with $P = 0.15$?

Solution

Step 1. H_0 : The number of mangled letters received by people follows a binomial distribution with $n = 2$, $p = 0.15$.

H_a : The number of mangled letters received by people doesn't follow a binomial distribution. With $n=2$, $P = 0.15$.

Step 2. $\alpha = 0.1$

$$V = K - 1 - m = 3 - 1 - 0 = 2$$

$$\chi^2_{\alpha, v} = \chi^2_{0.1, 2} = 4.61$$

Reject H_0 if sample $\chi^2 > 4.61$

Step 3. Sample χ^2

No. of mangled letters	Observed freq. (f_o)	Prob. with $n=2$ $P=0.15$	Expected freq ($f_e = np_i$)	$(f_o - f_e)^2$	$\frac{(f_o - f_e)^2}{f_e}$
0	260	0.7225	223.9750	1297.8006	5.7944
1	40	0.2550	79.0500	1524.9025	19.2904
2	10	0.0225	6.9750	9.1506	1.3119
$\sum \frac{(f_o - f_e)^2}{f_e}$					26.3967

Step 4. Reject H_0 . The number of hit over the game is not binomially distributed with $n = 2$ and $P = 0.15$.

4.4.2. Goodness-of-Fit Test for Poisson Distribution

Illustration One

It is hypothesized that the number of breakdowns per month of a computer system at a major university follows a Poisson distribution with $\mu = 2$. The data below show the observed number of breakdowns per month during a sample of 100 months. Use a 5% level of significance and test the null hypothesis.

Breakdowns	0	1	2	3	4	5 and above
Observed freq.	14	20	34	22	5	3

Solution

Step 1. H_0 : The population distribution of breakdowns is Poisson with $\mu = 2$.

H_a : The population distribution of breakdowns is not Poisson with $\mu = 2$.

Step 2. $\alpha = 0.05$

$$V = K - 1 - m = 6 - 1 - 0 = 5$$

$$\chi^2_{\alpha, v} = \chi^2_{0.05, 5} = 11.07$$

Reject H_0 if sample $\chi^2 > 11.07$

Step 3. Sample χ^2

Breakdowns	Observed freq. (f_o)	Prob. with $\lambda=2$	Expected freq (f_e $= np_i$)	($f_o - f_e$) ²	$\frac{(f_o - f_e)^2}{f_e}$
0	14	0.1353	13.53	0.2209	0.0163
1	20	0.2707	27.07	49.9849	1.8465
2	34	0.2707	27.07	48.0249	1.7741
3	22	0.1804	18.04	15.6816	0.8693
4	5	0.0902	9.02	16.1604	1.7916
5 or more	5	0.0527	5.27	0.0729	0.0138
$\sum \frac{(f_o - f_e)^2}{f_e}$					6.3117

Step 4. Do not Reject H_0 . The number of breakdowns per month of a computer system at the University follows a Poisson distribution with $\mu = 2$.

Illustration Two

Suppose that a teller supervisor believes that the distribution of random arrivals at a local bank is Poisson and sets out to test this hypothesis by gathering information. The following data represent a distribution of frequency of arrivals during one minute intervals at a bank. Use $\alpha = 0.05$ to test these data in an effort to determine whether they are Poisson distributed.

No. of arrivals	0	1	2	3	4	5 and above
Observed freq.	7	18	25	17	12	5

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Solution

Before we solve the question, first we have to compute the arrival rate per minute, and hence one degree of freedom is lost.

$$\lambda = \frac{\sum \left(\frac{\text{number of arrivals} *}{\text{observed frequency}} \right)}{\sum \text{Observed frequency}} = \frac{(0*7)+(18*1)+(25*2)+(17*3)+(12*4)+(5*5)}{84} = \frac{192}{84} = 2.3 \text{ cust / min}$$

Step 1. Ho: The arrival of customers at a bank is Poisson distributed with $\lambda = 2.3$

Ha: The arrival of customers at a bank is not Poisson distributed with $\lambda = 2.3$

Step 2. $\alpha = 0.05$

$$V = K - 1 - m = 6 - 1 - 1 = 4$$

$$\chi^2_{\alpha, v} = \chi^2_{0.05, 4} = 11.07$$

Reject Ho if sample $\chi^2 > 9.488$

Step 3. Sample χ^2

Number of arrivals	Observed freq. (f_o)	Prob. with $\lambda=2.3$	Expected freq ($f_e = np_i$)	($f_o - f_e$) ²	$\frac{(f_o - f_e)^2}{f_e}$
0	7	0.1003	8.4252	2.0312	0.2411
1	18	0.2306	19.3704	1.8778	0.0969
2	25	0.2652	22.2768	7.4158	0.3329
3	17	0.2033	17.0772	0.0060	0.0003
4	12	0.1169	9.8196	4.7541	0.4841
5 or more	5	0.0837	7.0308	4.1241	0.5866
$\sum \frac{(f_o - f_e)^2}{f_e}$					1.795

Step 4. Do not Reject Ho. The arrival of customers at a bank follows a Poisson distribution with $\lambda = 2.3$.

Illustration Three

The number of automobile accidents occurring per day in a particular city is believed to have a Poisson distribution. A sample of 80 days during the past year gives the data shown below. Do the data support the belief that the number of accidents per day has a Poisson distribution? Use $\alpha = 0.05$.

No. of accidents	0	1	2	3	4
Observed freq. (days)	3	25	11	7	3
	4				

Solution

Before we solve the question, first we have to compute the occurrence rate per day, and hence one degree of freedom is lost.

$$\lambda = \frac{\sum \left(\frac{\text{number of accidents} *}{\text{observed frequency}} \right)}{\sum \text{Observed frequency}} = \frac{(0 * 34) + (25 * 1) + (11 * 2) + (7 * 3) + (3 * 4)}{80} = \frac{80}{80} = 1 \text{ accident / day}$$

Step 1. Ho: The occurrence of accidents per day follows a Poisson dist. with $\lambda = 1.0$

Ha: The occurrence of accidents per day does not follow a Poisson dist. with $\lambda = 1.0$

Step 2. $\alpha = 0.05$

$$V = K - 1 - m = 4 - 1 - 1 = 2$$

$$\chi^2_{\alpha, v} = \chi^2_{0.05, 2} = 5.99$$

Reject Ho if sample $\chi^2 > 5.99$

Step 3. Sample χ^2

Number of accidents	Observed freq. (f_o)	Prob. with $\lambda=1.0$	Expected freq ($f_e = np_i$)	($f_o - f_e$) ²	$\frac{(f_o - f_e)^2}{f_e}$
0	34	0.3679	29.4320	20.8666	0.7090
1	25	0.3679	29.4320	19.6426	0.6674
2	11	0.1839	14.7120	13.7789	0.9366
3 or more	10	0.0803	6.4240	12.7878	1.9906
$\sum \frac{(f_o - f_e)^2}{f_e}$					4.3036

Step 4. Do not Reject Ho. The occurrence of accidents per day follows a Poisson distribution with $\lambda = 1.0$

4.4.3. Goodness-of-fit Test for Normal Distribution

Illustration One

Suppose that Ato Paulos developed an overall attitude scale to determine how his company's employees feel toward their company. In theory the scores can vary from 0 to 50. Ato Paulos pretests his measurement instrument on a randomly selected group of 100 employees. He tallies the scores and summarizes them into six categories as shown below. Are these pretest scores approximately normally distributed with $\mu = 24.9$ and $\sigma = 7.194$? Use $\alpha = 0.05$.

Score category	10-15	15-20	20-25	25-30	30-35	35-40
Frequency	11	14	24	28	13	10

Solution

Step 1. Ho: The attitude scores are normally distributed with $\mu = 24.9$ and $\sigma = 7.194$

Ha: The attitude scores are not normally distributed with $\mu = 24.9$ and $\sigma = 7.194$

Step 2. $\alpha = 0.05$

$$V = K - 1 - m = 6 - 1 - 0 = 5$$

$$\chi^2_{\alpha, v} = \chi^2_{0.05, 5} = 11.07$$

Reject Ho if sample $\chi^2 > 11.07$

Step 3. Sample χ^2

$$Z = \frac{X - \mu}{\sigma}$$

With $Z = \frac{X - \mu}{\sigma}$, the expected probability of each category can be obtained as follows:

For category 10-15		Probability
	$z_{10} = \frac{10 - 24.9}{7.194} = -2.07$	0.48077
	$z_{15} = \frac{15 - 24.9}{7.194} = -1.38$	- 0.41621
Expected probability		0.06456
For category 15-20		Probability
	$z_{15} = \frac{15 - 24.9}{7.194} = -1.38$	0.41621
	$z_{20} = \frac{20 - 24.9}{7.194} = -0.68$	- 0.25175
Expected probability		0.16446
For category 20-25		Probability
	$z_{20} = \frac{20 - 24.9}{7.194} = -0.68$	0.25175
	$z_{25} = \frac{25 - 24.9}{7.194} = +0.01$	+0.00399
Expected probability		0.25574
For category 25-30		Probability
	$z_{25} = \frac{25 - 24.9}{7.194} = +0.01$	- 0.00399
	$z_{30} = \frac{30 - 24.9}{7.194} = +0.71$	+0.26115
Expected probability		0.25716
For category 30-35		Probability
	$z_{30} = \frac{30 - 24.9}{7.194} = +0.71$	- 0.26115
	$z_{35} = \frac{35 - 24.9}{7.194} = +1.40$	0.41924
Expected probability		0.15809
For category 35-40		Probability
	$z_{35} = \frac{35 - 24.9}{7.194} = +1.40$	- 0.41924
	$z_{40} = \frac{40 - 24.9}{7.194} = +2.10$	0.48214
Expected probability		0.06290

The six probabilities do not sum to 1.00. Even though observed frequencies were obtained only for these six categories, getting a score less than 10 or greater than 40 was also possible. Because 0.5 of the probabilities lie in each half of a normal distribution and utilizing the sum of expected probabilities on each side of the mean, 24.9, we can obtain a probability of the < 10 category: $0.5 - (0.06456 + 0.16446 + 0.25175) = 0.01923$. Similarly, we can obtain the probability of >40 category: $0.5 - (0.00399 + 0.25716 + 0.15809 + 0.06290) = 0.01786$. Expected frequencies can then be obtained by multiplying each expected probability by the total frequency (100), as shown below.

Score category	Probability	Expected freq (fe = np _i)	
< 10	0.01923	1.923	8.379
10 – 15	0.06456	6.456	
15 – 20	0.16446	16.446	16.446

20 – 25	0.25574	25.574	25.574
25 – 30	0.25716	25.716	25.716
30 -35	0.15809	15.809	15.809
35 – 40	0.06290	6.290	8.076
> 40	0.01786	1.786	

As the < 10 and > 40 categories have values of less than 5, each must be combined with the adjacent category. As a result, the < 10 category becomes part of the 10 – 15 category and the > 40 category becomes part of the 35 – 40 category.

Score category	Probability	Expected freq (fe = np _i)
10 – 15	0.08379	8.379
15 – 20	0.16446	16.446
20 – 25	0.25574	25.574
25 – 30	0.25716	25.716
30 -35	0.15809	15.809
35 – 40	0.08076	8.076

The value of the chi-square can then be computed.

Score category	Observed freq. (f _o)	Probability	Expected freq (fe = np _i)	(fo-fe) ²	$\frac{(f_o - f_e)^2}{f_e}$
10 – 15	11	0.08379	8.379	6.8696	0.8199
15 – 20	14	0.16446	16.446	5.9829	0.3638
20 – 25	24	0.25574	25.574	2.4775	0.0964
25 – 30	28	0.25716	25.716	5.2167	0.2029
30 -35	13	0.15809	15.809	7.8905	0.4991
35 – 40	10	0.08076	8.076	3.7018	0.4584
$\sum \frac{(f_o - f_e)^2}{f_e}$					2.4409

Step 4. Do not Reject H_o. The attitude score are normally distributed with mean 24.9 and standard deviation 7.194.

Illustration Two

The director of a major soccer team believes that the ages of purchasers of game tickets are normally distributed. If the following data represent the distribution of ages for a sample of observed purchasers of major soccer game tickets, use the chi-square goodness-of-fit test to determine whether this distribution is significantly different from the normal distribution. Assume that $\alpha = 0.05$.

Age of purchaser	10-20	20-30	30-40	40-50	50-60	60-70
Frequency	16	44	61	56	35	19

Solution

Step 1. Ho: The ages of purchasers of soccer game tickets are normally distributed.

Ha: The ages of purchasers of soccer game tickets aren't normally distributed

Step 2. $\alpha = 0.05$

$$V = K - 1 - 2 = 6 - 1 - 2 = 3$$

$$\chi^2_{\alpha, v} = \chi^2_{0.05, 3} = 7.81$$

Reject Ho if sample $\chi^2 > 7.81$

Step 3. Sample χ^2

Age category	Observed freq	Mid point (M)	fm	fm ²
10-20	16	15	240	3,600
20-30	44	25	1,100	27,500
30-40	61	35	2,135	74,725
40-50	56	45	2,520	113,400
50-60	35	55	1,925	105,875
60-70	19	65	1,235	80,275
	231	$\sum fm = 9,155$		$\sum fm^2 = 405,375$

$$\bar{X} = \frac{\sum fm}{n} = \frac{9,155}{231} = 39.63$$

$$S = \sqrt{\frac{\sum fm^2 - \frac{(\sum fm)^2}{n}}{n-1}} = \sqrt{\frac{405,375 - \frac{(9,155)^2}{231}}{231-1}} = 13.60$$

$$Z = \frac{X - \mu}{\sigma}$$

With $\frac{X - \mu}{\sigma}$, the expected probability of each category can be obtained as follows:

For category 10-20

Probability

$$z_{10} = \frac{10 - 39.63}{13.6} = -2.18$$

0.48537

$$z_{20} = \frac{20 - 39.63}{13.6} = -1.44 \quad - 0.42507$$

Expected probability **0.06030**

For category 20-30 **Probability**

$$z_{20} = \frac{20 - 39.63}{13.6} = -1.44 \quad 0.42507$$

$$z_{30} = \frac{30 - 39.63}{13.6} = -0.71 \quad - 0.26115$$

Expected probability **0.16392**

For category 30-45 **Probability**

$$z_{30} = \frac{30 - 39.63}{13.6} = -0.71 \quad 0.26115$$

$$z_{40} = \frac{40 - 39.63}{13.6} = +0.03 \quad + 0.01197$$

Expected probability **0.27312**

For category 40-50 **Probability**

$$z_{50} = \frac{50 - 39.63}{13.6} = +0.76 \quad 0.27637$$

$$z_{40} = \frac{40 - 39.63}{13.6} = +0.03 \quad - 0.01197$$

Expected probability **0.26440**

For category 50-60 **Probability**

$$z_{60} = \frac{60 - 39.63}{13.6} = +1.50 \quad 0.43319$$

$$z_{50} = \frac{50 - 39.63}{13.6} = +0.76 \quad - 0.27637$$

Expected probability **0.15682**

For category 60-70 **Probability**

$$z_{70} = \frac{70 - 39.63}{13.6} = +2.23 \quad 0.48713$$

$$z_{60} = \frac{60 - 39.63}{13.6} = +1.50$$

$$- 0.43319$$

Expected probability

0.05394

The six probabilities do not sum to 1.00. Even though observed frequencies were obtained only for these six categories, getting a score less than 10 or greater than 70 is also possible.

For < 10:

Probability between 10 and the mean = $0.06030 + 0.16392 + 0.26115 = 0.48537$.

Probability < 10 = $0.5 - 0.48537 = 0.01463$

For > 70

Probability between 70 and the mean = $0.05394 + 0.15682 + 0.2640 + 0.01197 = 0.48713$

Probability > 70 = $0.5 - 0.48713 = 0.01287$

Then, the expected frequencies can be obtained by multiplying each expected probability by the total frequency (231) as follows:

Age category	Probability	Expected freq (fe = np _i)	
< 10	0.01463	3.380	17.309
10 – 20	0.06030	13.929	
20 – 30	0.16392	37.866	37.866
30 – 40	0.27312	63.091	63.091
40 – 50	0.26440	61.076	61.076
50 – 60	0.15682	36.225	36.225
60 – 70	0.05394	12.460	15.433
> 70	0.01287	2.973	

Since the < 10 and > 70 categories have values of less than 5, each must be combined with the adjacent category. As a result, the < 10 category becomes part of the 10 – 20 category and the > 70 category becomes part of the 60 – 70 category.

Age category	Probability	Expected freq (fe = np _i)
10 – 20	0.07493	17.309
20 – 30	0.16392	37.866
30 – 40	0.27312	63.091
40 – 50	0.26440	61.076
50 – 60	0.15682	36.225
60 – 70	0.06681	15.433

The value of the chi-square can then be computed.

Score category	Observed freq. (f_o)	Probability	Expected freq ($f_e = np_i$)	$(f_o - f_e)^2$	$\frac{(f_o - f_e)^2}{f_e}$
10 – 20	16	0.07493	17.309	1.7135	0.0990
20 – 30	44	0.16392	37.866	37.6260	0.9937
30 – 40	61	0.27312	63.091	4.3723	0.0693
40 – 50	56	0.26440	61.076	25.7658	0.4219
50 – 60	35	0.15682	36.225	1.5006	0.0414
60 – 70	19	0.06681	15.433	12.7235	0.8244
$\sum \frac{(f_o - f_e)^2}{f_e}$					2.4497

Step 4. Do not Reject H_o . The age of purchasers of soccer game tickets are normally distributed.